

Doc. 310 – Resonance Geometry as the Overarching Reference

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Doc. 310 – Resonance Geometry as the Overarching Reference

What this is about

The recurring objection to FFGFT is: “ ξ is fitted to the lepton masses, hence circular.” This note shows that the objection rests on a misunderstanding of the hierarchy.

The correct order is: **overarching** is the geometry that possesses resonances — dimensionless, fixing all ratios. The **anchor** is a single measured value — carrying no structure, only the scale. The measured value is the tuning pitch, not the music.

Not part of the A-series.

Conventions

Legend.

- [K]** *Core derivation* — derived from the foundational posits, reproducible.
- [S]** *Structural claim* — justified, not fully proved.
- [POSIT]** *Posit* — declared assumption.

Central quantities.

Q	Koide ratio $(\sum m)/(\sum \sqrt{m})^2$; measures $2/3$ [K].
$\xi = 4/30000$	Fundamental parameter; form [K], magnitude 5^4 [POSIT, P33].
r_ℓ, p_ℓ	Prefactors and exponents of the mass ladder (Doc. 190).
v	Vacuum expectation value; sets the absolute scale.
$S^2 = 4\pi, S^3 = 2\pi^2$	Sphere surfaces of the T^4 boundary geometry.
μ_0, ϵ_0	Vacuum constants; follow from α, e, c (Doc. 013).

Chapter A

The Three Levels

Level	Content	Status
0 — Measurement	$m_\mu/m_e = 206.77, m_\tau/m_e = 3477$	measured
1 — Relation	$Q = 2/3 \text{ to } 10^{-5}$	parameter-free [K]
2 — Encoding	$m_\ell = r_\ell \xi^{p_\ell} \nu$	notation

The only error would be to sell Level 2 as Level 0 — to claim ξ is the measured starting point. As long as each level's status is declared, the account is both honest and transparent.

Chapter B

What the Geometry Fixes

The resonance structure makes three fully dimensionless statements:

(1) Koide relation [K]. The three $\sqrt{m}$ form an angle of 44.9997° to the vector $(1, 1, 1)$:

$$Q = \frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} = \frac{2}{3}.$$

No measured value needed — only the ratios of the masses.

(2) Generation ladder [K]. The charged leptons realize the exponents $p_e = 3/2$, $p_\mu = 1$, $p_\tau = 2/3$ — a **geometric sequence** with constant ratio $q = 2/3$:

$$p_{n+1} = \frac{2}{3} p_n, \quad \frac{3}{2} \xrightarrow{\times 2/3} 1 \xrightarrow{\times 2/3} \frac{2}{3}.$$

The ratio $q = 2/3$ is the same value as the Koide value $Q = 2/3$ — the same number orders the ladder and the mass relation (Doc. 258). The sequence is verifiable against the three measured masses; that other sectors sit on other rungs does not affect this.

(3) Harmonic prefactors [POSIT]. The prefactors $r_e = 4/3$, $r_\mu = 16/5$, $r_\tau = 25/9 = (5/3)^2$ — intervals in the Euler Tonnetz; a derivation of the r_i from an (n, l, j) formula does not hold, they remain posits.

These statements fix the ratios to a few digits (`ffgft_310_massenleiter.py`):

$$m_\mu/m_e = 207.85 \text{ (+0.52\%)}, \quad m_\tau/m_e = 3531.6 \text{ (+1.56\%)}. \quad \square$$

Chapter C

Where the Measured Value Enters

Only for the **absolute scale** is an anchor needed: $m_e = 0.511$ MeV. It does not say how the masses behave — the geometry does. It says only: “this structure, measured in kg, starts here.” One could equally take m_μ or m_τ — the ratios would be identical.

The music analogy.

Music	Leptons
Intervals (2:1, 3:2, 5:4)	$Q = 2/3$, 1/3-ladder, harm. r
Tuning pitch $a' = 440$ Hz	$m_e = 0.511$ MeV

One can choose $a' = 440$ or 442 Hz — the music stays the same. The anchor is convention, the resonance structure is the overarching thing.

Chapter D

Why This Resolves the Circularity

The objection: “ ξ is fitted to the masses.” The answer: yes — the absolute *scale* (the tuning pitch) is measured. But the *structure* (the resonances) is not fitted. It is **confirmed** by $Q = 2/3$, not set.

The proof is in the residual (`ffgft_310_koide_referenz.py`):

$$Q_{\text{measured}} - \frac{2}{3} = -6.2 \times 10^{-6} \neq 0.$$

A fit would have hit Q exactly (0.000%). The geometry hits Q to 10^{-5} *with a residual* — the signature of a genuine prediction.

Chapter E

The Choice of Starting Point Is Convention

There is **one** dimensionless number world. $\xi, \alpha, Q, m_\mu/m_e$ are different coordinates in it — none more “fundamental.” The circularity is not ξ -specific: “why $\alpha = 1/137$?”, “why $Q = 2/3$?”, “why $\xi = 1/7500$?” are all open. Every theory must set some dimensionless number somewhere.

The only real difference is **one networked posit vs. many isolated ones**: the SM has 3 independent lepton masses; FFGFT saves one number via $Q = 2/3$ and connects, via ξ , the lepton sector to cosmology (a_0, H_0). The gain is real but modest — not “everything from nothing,” but a reduction with direction.

Chapter F

The Discovery Path and the Geometry of the π Factors

The corpus presents $\xi \rightarrow$ masses $\rightarrow$ cosmology (tidied up). The actual discovery path was: (1) QM/RT incompatibility as motive, (2) the Higgs formula as first expression, (3) vacuum and its constants, (4) only then masses, Koide, Tonnetz.

F.1 Higgs formula as a second independent access [K]

$$\xi = \frac{m_h^2}{64\pi^3 v^2} = 1.30 \times 10^{-4}$$

lies about 2.3 % below $\xi = 4/30000$ (2.2–2.6 % depending on inputs), from entirely different inputs (Higgs mass, vacuum value) than the lepton masses (ffgft_310_higgs_zugang.py). Two independent sectors, one ξ — that was the signal. The Higgs formula is thus a consistency check of ξ .

F.2 π^3 is geometry, not a loop [K]

The factor π^3 is *not* a quantum loop factor but the product of the sphere surfaces of the compact T^4 boundary geometry (ffgft_310_pi_geometrie.py):

$$16\pi^3 = (4\pi) \cdot (4\pi^2) = S^2 \cdot 2S^3.$$

The earlier version $64\pi^4$ contained a surplus factor $64\pi^4/16\pi^3 = 4\pi$. Its origin is documented (Doc. 385): the original vacuum formula read $\lambda_h^2 v^2 e^2 / (64\pi^4 \epsilon_0 \hbar c m_h^2) = \xi_{\text{EFT}} \cdot \alpha$; the 4π is the one from $4\pi\epsilon_0$ in $\alpha = e^2 / (4\pi\epsilon_0 \hbar c)$ and was not cancelled when ϵ_0 was dropped (transition to $\alpha = 1$). Geometrically, $4\pi = S^2$ can be read as **one 2-sphere too many**; the error itself, however, is a leftover of units and not an independent confirmation of the geometric counting.

F.3 The repeated μ_0/ϵ_0 structure [K]

In the origin documents (Docs. 011, 013, 015) μ_0 and ϵ_0 are *not* independent constants — they follow from α, e, c :

$$\epsilon_0 = \frac{e^2}{2\alpha\hbar c}, \quad \mu_0 = \frac{2\alpha\hbar}{e^2 c}, \quad \epsilon_0 \mu_0 c^2 = 1.$$

They are where geometry and units become separately visible: $\mu_0 = \underbrace{4\pi}_{\text{geometry } S^2} \times \underbrace{10^{-7}}_{\text{unit}}$. In the α formula the 4π cancels (exactly in the SI before 2019, today to 10^{-10}):

$$\alpha = \frac{e^2 \mu_0 c}{4\pi \hbar} \approx \frac{e^2 \times 10^{-7} \times c}{\hbar}.$$

Since the 2019 SI revision μ_0 is a measured value (CODATA 2022: $1.25663706127(20) \cdot 10^{-6} \text{ N/A}^2$); $\mu_0/(4\pi \cdot 10^{-7}) - 1 = -1.3 \cdot 10^{-10}$, within 1σ — numerically negligible. Precisely this 4π from $4\pi\epsilon_0$ is the origin of the earlier Higgs prefactor $64\pi^4 = 16\pi^3 \cdot 4\pi$ (see the previous section and Doc. 385). Geometry against geometry. In natural units ($\hbar = c = 1$) μ_0, ϵ_0 vanish entirely; α and ξ remain as numbers. Setting additionally $\alpha = 1$ (T0 units, Doc. 011) redefines the charge — $\alpha = e^2 = 1 \Rightarrow e = 1$ — and α vanishes too. Then **only** ξ remains as the single free dimensionless parameter.

Conclusion: Nothing was fitted to the target. The π factors are throughout sphere surfaces of the field geometry; a clean unit and dimension count yields them necessarily.

Chapter G

Why Mixing QM and RT Is Legitimate

The usual objection — “you mix $\hbar$ (quantum) with c (relativity)” — is not defended but inverted: the **separation** QM|RT is the artificial thing, not the mixing.

(1) $\hbar$ and c are both only unit bridges (space $\leftrightarrow$ time, energy $\leftrightarrow$ time). Neither is a theory. Using both at once is permitted.

(2) The T^4 structure needs both (P34): c is the space-time bridge, $\hbar$ the mass-time bridge. Setting $c = \hbar = 1$ collapses the geometry. To see it, one must mix.

(3) $\tilde{T} \cdot m = 1$ is the mixing: proper time (RT) times mass (QM). Physics has long mixed them anyway — Dirac, QFT, Unruh, Hawking. FFGFT only makes it explicit.

(4) The conflict is one of interpretation, not formalism. Per Doc. 230, instantaneity is a description artefact; on T^4 Bell correlations are topologically connected, not ontologically non-local. Correct the interpretation and mixing becomes necessary.

Core statement. To forbid the mixing is to forbid the geometry.

Chapter H

Core Statement

The overarching thing is the geometry that possesses resonances. Its dimensionless statements — $Q = 2/3$, the generation ladder (exponents $3/2, 1, 2/3$), the harmonic prefactors — fix the ratios to a few digits. The measured value (one mass in SI) is only the tuning pitch: it says *where* the structure sounds, not *how* it sounds.

Chapter I

References

Internal documents: Doc. 011 (fine structure, μ_0/ε_0), Doc. 013 (SI, μ_0/ε_0 from $\alpha/e/c$), Doc. 015 (natural units), Doc. 049 (Higgs integration), Doc. 060 (musical resonance), Doc. 158 (Koide-to-g-2 bridge), Doc. 190 (P33, P41–P43, mass formula 190.3, Higgs formula 190.1), Doc. 230 (Bell on T^4), Doc. 306 ($T \cdot E = 1$), Doc. 309 (scale anchor).

Verification scripts (no free parameters, all inputs declared):
ffgft_310_koide_referenz.py, ffgft_310_massenleiter.py,
ffgft_310_pi_geometrie.py, ffgft_310_higgs_zugang.py.